Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/12/2005
PI Name:	Javanainen, Juha D.Sc.		
Project Title:	Quantum Optics Style Theory for Degenerate Bose and Fermi Gases		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06269-3046	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007, per N. Hall-GRC (9/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2880		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In the broadest of terms, we are currently pursuing two lines of research. First, we are developing a quantum optics style theory for dilute quantum degenerate Bose and Fermi gases in the neighborhood of a Feshabch resonance. Second, we are studying the dynamics of ultralow-temperature gases in optical lattices. Our goal is to generate new theoretical concepts in the physics of degenerate atomic gases, such as a Bose-Einstein condensate or a degenerate Fermi gas realized in an alkali metal vapor. In the usual manner of basic theoretical research, when the concepts and ideas arising from this work circulate in the community, new theoretical ideas, experiments, and ultimately new technologies will follow. Prospective applications of interest to NASA include atomic clocks and deep-space navigation.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Conversion of a pair of atoms into a diatomic molecule close to a Feshbach resonance, the proximity being controlled by a magnetic field applied to the gas sample, is presently one of the most active frontiers in Atomic, Molecular, and Optical physics. We have materially completed two projects in this area: 1. We have formulated a simple mean-field theory for the atom-molecule conversion in a zero-temperature Fermi gas. We start from the dynamical equations of motion under the assumption that the molecules are treated as a classical field analogous to laser light, solve the steady state with the assumption that atoms only appear and disappear as a result of dissociation of, or association into, molecules, and implement the theory on a computer. We calculate the fraction of atoms converted into molecules as a function of the magnetic field, and find a good agreement with the experiments [Partridge et al., Phys. Rev. Lett. 95, 020404 (2005)] over a dynamical range of five orders of magnitude. Our calculations are at variance with common views in the field in two ways. First, we insist that the molecules have to be considered explicitly, not hidden away into an atom-atom scattering length. Second, the results depend on the detailed structure of the molecules even in the case of strong atom-molecule coupling, where one might expect structure-independent universal results. 2. We have studied the atom-molecule conversion efficiency in a nondegenerate, i.e., ordinary, thermal gas when a magnetic field is swept across the Feshbach resonance . Starting directly from the time dependent Schroedinger equation, we have shown that the conversion efficiency depends on the density and the temperature of the gas only through a quantity known as phase space density. This result, too, agrees quantitatively with experimental findings [Hodby et al., Phys. Rev. Lett. 94, 120402 (2005)]. We have also worked on several incidental projects. We have studied an atom with a level scheme that supports nonabsorbing (dark) states under a light field whose polarization fluctuates, and found that the dynamics rather depends on the fluctuations of the light than the usual variables such as intensity. We have added detail to our earlier analysis of Feshbach resonance in a system that only includes atomic and molecular Bose-Einstein condensates. Finally, we have written a solicited Perspective article for Science on a pathbreaking experiment of the MIT group headed by Wolfgang Ketterle.
Bibliography Type:	Description: (Last Updated: 01/08/2020)
Articles in Peer-reviewed Journals	Mackie M, Collin A, Javanainen J. "Comment on “Stimulated Raman adiabatic passage from an atomic to a molecular Bose-Einstein condensate." Physical Review A. 2005 Jan;71(1):017601. http://dx.doi.org/10.1103/PhysRevA.71.017601 , Jan-2005
Presentation	Javanainen, J. "Molecules out of a Fermi gas at a Feshbach resonance: A simple approach" Invited talk at the 2005 Obergurgl meeting, Obergurgl, Austria, 2/27/05 – 3/5/05 Mar-2005
Presentation	Javanainen, J.; Carmichael, A. "A simple approach to BEC-BCS crossover at a Feshbach resonance in a degenerate Fermi gas" Contributed talk in DAMOP 2005, Lincoln, NE, 5/17-5/21, 2005. May-2005
Presentation	Carmichael, A.; Javanainen, J.; Kostrun. M. "Some new results on the two-well Bose-Hubbard model" Poster in DAMOP 2005, Lincoln, NE, 5/17-5/21, 2005. May-2005